Unit 3.2 Printables: Age of the Earth

Standard: HS-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. Targets are labeled ESS1-5.1 to ESS1-5.5 (priority) and ESS1-6.1 to ESS1-6.3 (supporting). Pacing follows the 26-27 workbook: 12 days across 10 lessons (Lessons 7 and 8 each take two days).

Everything the class needs for this unit is in this one file, organized by lesson. Print on white paper in black ink only. "Copies" says how many to print: per student, per pair, per group or one for the class. Each item is a dashed box; cut cards along the dashed lines. Sources are cited on any item that uses real data; items marked illustrative use simplified numbers made up for class.

Lesson	Printable	Use	Copies
1	Pre-check (not graded)	Glue under "Pre-check" in the notebook; keep to compare at the end of the unit	One per student
1	Hook data: ocean floor and continent ages	Glue under "Hook data" in the notebook (D1-D)	One per student
1	Unit CER page	Glue into the notebook (U-1); add rows all unit	One per student
1	World map and site list	Glue under "Plot the pattern" in the notebook (D1-A)	One per student
1	Frayer card	Make one for each of the 8 vocabulary words as they come up; finish by Lesson 9	8 per student
2	Boundary table	Glue under "Plate boundary models" in the notebook (D1-B)	One per student
2	Plate boundary word wall: 8 vocabulary cards	Cut apart; pairs match word, definition and picture	One set per pair
3	Seafloor ages: table, color key and 13-box strip	Glue under "Seafloor ages" and "Color the seafloor" (D2-A, D2-B)	One per student
3	Graph grid: age versus distance	Glue under "Graph: age vs. distance" (D2-C)	One per student
3	North America ages: table and 5-box strip	Glue under "Color North America" (D2-D)	One per student
4	45 cm Earth timeline: event table and three strips	Glue under "45 cm Earth timeline" (D3-B); cut and tape the strips end to end	One per student
4	Early Earth evidence log	Glue under "Evidence list" in the notebook	One per student
5	Crater counts: record sheet	Glue under "Crater counts" in the notebook (D3-A)	One per student
5	Half-life model: decay record	Quick lab (textbook p. 215); glue under "Half-life" in the notebook	One per pair
6	Guided notes: the life of ocean crust and how continents grow	Glue under "Guided notes" in the notebook (D4-A)	One per student
6	Density lab table	Glue under "Density lab table" in the notebook (D4-B)	One per student
6	SketchNotes checklist	Glue above your half page of notes (D4-C)	One per student
7	Magnetic stripes: band table and paper strips	Day 1. Glue under "Magnetic stripes" (D5-A); cut and fold the strips	One per pair
7	Hawaii and GPS: measured motion	Day 1. Glue under "Hawaii and GPS" (D5-B)	One per student
7	Pangaea puzzle: continent cutouts and fossil table	Day 2. Cut out the continents; glue the fossil table under "Pangaea puzzle" (D5-C)	One per pair
7	Evaluate the evidence table	Day 2. Glue under "Evaluate the evidence" (D5-D)	One per student

8	Three buckets: sort your evidence	Day 1. Glue under "Sort your evidence" (D6-E)	One per student
8	Part A sentence frames and data bank	Day 1. Glue under "Part A: argument" (D6-A)	One per student
8	Star and step peer feedback	Day 1. Glue under "Peer feedback" (D6-D)	One per student
8	Earth's story: evidence cards and frames	Day 2. Cut out the cards; glue under "Part B" (D6-B, D6-C)	One per student
9	Self-rating chart and work plan	Glue under "Self-rating chart" in the notebook (D7-A)	One per student
9	Review stations: five station cards	Cut apart; early finishers rotate through the stations	One set per station table

Lessons with no printable (10) use the notebook and the slides only.

UNIT 3.2 · LESSON 1 · PRE-ASSESSMENT

Pre-check (not graded)

Use: Glue under "Pre-check" in the notebook; keep to compare at the end of the unit · Copies: One per student

Answer in your notebook. It is OK to guess. We will come back to these at the end of the unit.

1. About how old is Earth? _____

2. What is the oldest rock or mineral that has been found on Earth, about how old is it, and where was it found?
_____

3. Is the ocean floor older, younger, or about the same age as the continents? What is your reason?
_____

_____

_____

_____

4. What do you think the Earth's outer layer is made of, and does it ever move? _____

_____

_____

5. How could scientists tell how old a rock is? _____

_____

_____

UNIT 3.2 · LESSON 1 · HOOK DATA

Hook data: ocean floor and continent ages

Use: Glue under "Hook data" in the notebook (D1-D) · Copies: One per student

The ocean floor

Place on the ocean floor	Age of the rock
Right at the Mid-Atlantic Ridge	0 (forming now)
Middle of the Atlantic, 2,000 km from the ridge	about 80 million years
Atlantic floor next to eastern North America	about 180 million years
Western Pacific, east of Japan	about 180 million years

Ages are rounded. Earth is about 4,500 million years old.

The continents and life

Continental material	Where	Age
Oldest mineral (zircon)	Western Australia	about 4.4 billion years
Oldest known rock	Canada	about 4.0 billion years
Oldest fossils	Western Australia	about 3.5 billion years

I notice: _____

_____

_____

I wonder: _____

_____

_____

Sources: ocean-floor ages, Müller et al. (2008), Science 319:1357-1362, "Age, spreading rates, and spreading asymmetry of the world's ocean crust" (oldest ocean floor about 180 million years; rounded); oldest mineral (zircon), Wilde et al. (2001), Nature 409:175-178 (about 4.4 billion years, Jack Hills, Western Australia); oldest known rock, Bowring and Williams (1999), Contributions to Mineralogy and Petrology 134:3-16 (Acasta Gneiss, Canada, about 4.0 billion years); oldest fossils, Allwood et al. (2006), Nature 441:714-718 (about 3.5 billion years, Western Australia).

UNIT 3.2 · LESSON 1 · UNIT CER

Unit CER page

Use: Glue into the notebook (U-1); add rows all unit · Copies: One per student

Write your first claim today. It is fine if it is wrong. Revise it every few lessons and add evidence to the log.

First claim (Lesson 1): Why is the seafloor so much younger than the continents?

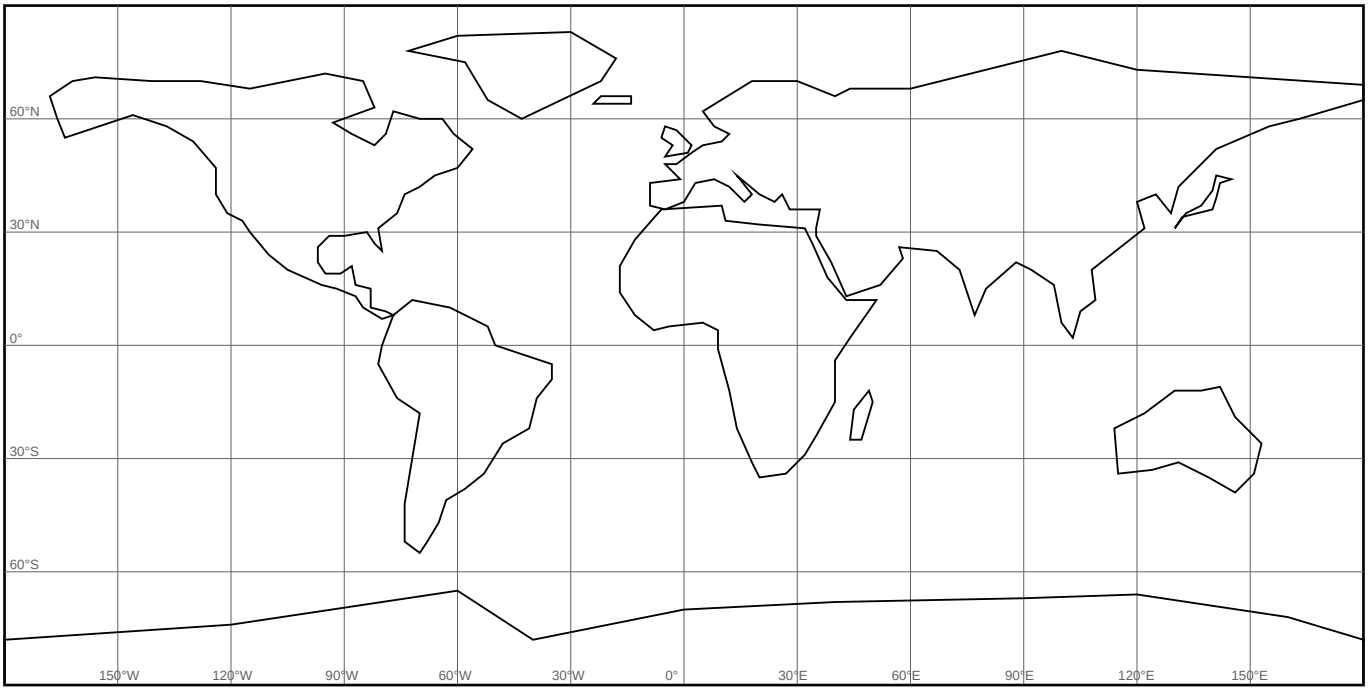
Evidence (what I saw, with a number)	Lesson	Kind (map, graph, data, model, fossil)	What it shows

Revised claim (Lesson 8)

World map and site list

Use: Glue under "Plot the pattern" in the notebook (D1-A) · Copies: One per student

Lab A. Each partner plots 5 sites. Use a red dot for a volcano and a blue X for an earthquake. The lines are 30° apart. Do the points make lines, or are they random? Circle the one site that does not fit.



Site	Type	Latitude, longitude
Mount St. Helens, Washington, USA	volcano	46.2° N, 122.2° W
Mount Fuji, Japan	volcano	35.4° N, 138.7° E
Popocatépetl, Mexico	volcano	19.0° N, 98.6° W
Hekla, Iceland	volcano	64.0° N, 19.7° W
Erta Ale, Ethiopia	volcano	13.6° N, 40.7° E
Tohoku, Japan (2011, magnitude 9.1)	earthquake	38.3° N, 142.4° E
Maule, Chile (2010, magnitude 8.8)	earthquake	35.9° S, 72.7° W
Haiti (2010, magnitude 7.0)	earthquake	18.4° N, 72.5° W
Turkey (2023, magnitude 7.8)	earthquake	37.2° N, 37.0° E
Kilauea, Hawai'i, USA	volcano	19.4° N, 155.3° W

The points make _____ . They match the edges of the _____ .

Coordinates: Smithsonian Global Volcanism Program and the USGS Earthquake Hazards Program catalogs (rounded). Continent outlines are a simplified schematic map, not for measuring distances.

UNIT 3.2 · LESSON 1 · VOCABULARY

Frayer card

Use: Make one for each of the 8 vocabulary words as they come up; finish by Lesson 9 · Copies: 8 per student

Definition (my words)	Characteristics

WORD: _____

Examples	Non-examples

Words: lithosphere (plate), plate boundary, mid-ocean ridge, subduction, trench, magnetic reversal, craton, impact crater.

Boundary table

Use: Glue under "Plate boundary models" in the notebook (D1-B) · Copies: One per student

Model each boundary with clay (about 5 minutes each). Sketch each model in your notebook and label the arrows. Then fill in the table.

Boundary	What the plates did	What formed	Real example
Divergent			
Convergent			
Transform			

Where is ocean crust made? _____ Where is it destroyed? _____

Boundary sketches (label the arrows)

Divergent

Convergent (trench, volcanoes)

Transform

Examples are standard textbook cases (Mid-Atlantic Ridge, Andes and Japan trenches, San Andreas Fault).

UNIT 3.2 · LESSON 2 · VOCABULARY

Plate boundary word wall: 8 vocabulary cards

Use: Cut apart; pairs match word, definition and picture · Copies: One set per pair

lithosphere (plate) · Earth's rigid outer layer	**plate boundary · where two plates meet**
mid-ocean ridge · plates move apart; new crust	**subduction · one plate sinks under another**
trench · deep groove at a sinking plate	**magnetic reversal · the magnetic field flips**
craton · a continent's ancient core	**impact crater · hole made by a space rock**

Word on the left of each card, definition on the right. Cut on the dashed lines and shuffle. Match words to definitions, then sketch a picture for each word.

Seafloor ages: table, color key and 13-box strip

Use: Glue under "Seafloor ages" and "Color the seafloor" (D2-A, D2-B) · Copies: One per student

Lab A. Draw a row of 13 boxes: the middle box is the RIDGE, and the others are 500, 1,000 ... 3,000 km on each side. Color each box with the key.

Distance from ridge (km)	Crust age (million years)	Color
0	0	red
500	20	orange
1,000	40	yellow
1,500	60	green
2,000	80	blue
2,500	100	purple
3,000	120	dark purple

The strip (write the distance in each box, then color):

Where is the youngest color? The oldest? How do the left and right sides compare? As you move away from the ridge, the crust gets _____ .

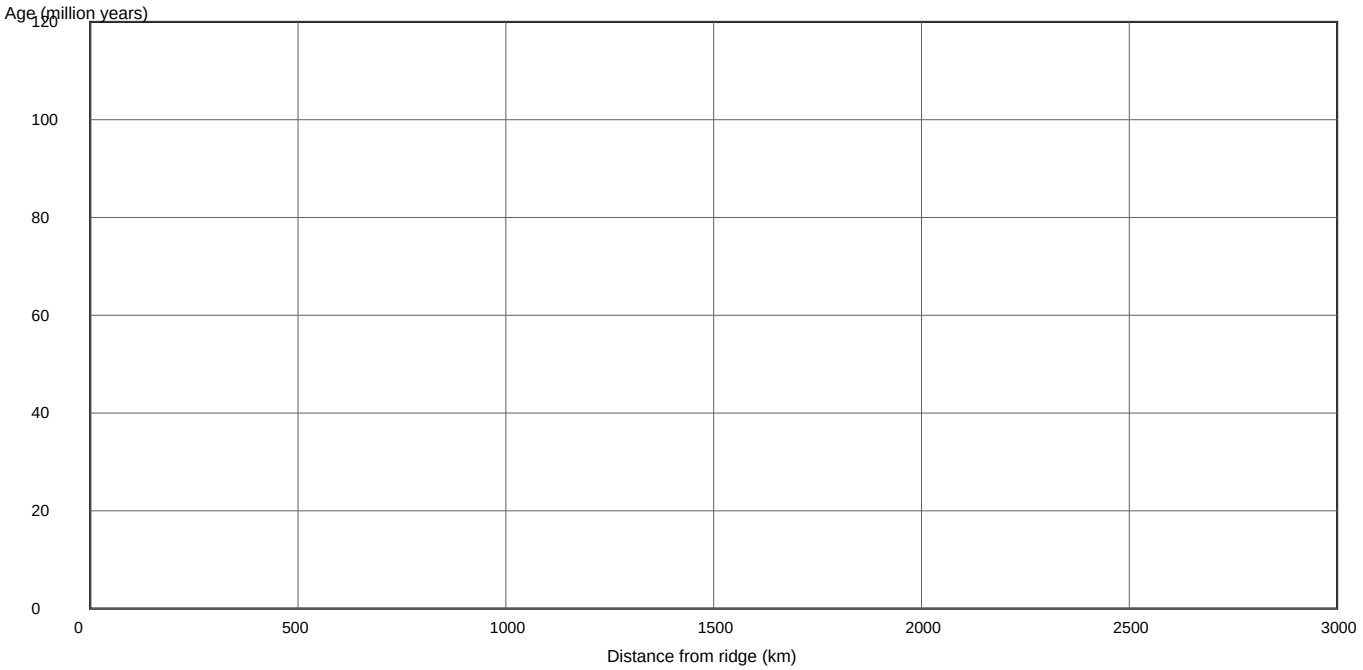
The spreading rate (2.5 cm per year, or 25 km per million years) is close to the Atlantic half-rate; see Müller et al. (2008), Science 319:1357-1362. Values are simplified.

UNIT 3.2 · LESSON 3 · GRAPH

Graph grid: age versus distance

Use: Glue under "Graph: age vs. distance" (D2-C) · Copies: One per student

Lab B. x-axis: distance from the ridge (0 to 3,000 km). y-axis: age (0 to 120 million years). Plot the 7 points from the table and connect them.



Pick two points. Rate = distance ÷ time (500 km in 20 million years). Show your work: _____

Rate: _____ km per million years = _____ cm per year (1 km per million years = 0.1 cm per year). About how fast do fingernails grow? _____

Use your line to predict the age of crust 3,500 km from the ridge: _____ million years.

North America ages: table and 5-box strip

Use: Glue under "Color North America" (D2-D) · Copies: One per student

Lab C. Draw a row of 5 boxes from west to east. Write the region names in order. Oldest region: dark brown. Youngest: light yellow. Pick colors in between.

Region	From the core	Age of the crust
Pacific Coast Ranges	west, 2,800 km	about 50 million years
Rocky Mountain region	west, 1,800 km	about 150 million years
Canadian Shield (core)	0	about 2.7 billion years
Grenville Province	east, 1,200 km	about 1.1 billion years
Appalachian Mountains	east, 2,000 km	about 400 million years

Where is the oldest crust? How does the age change as you move away from it? _____

Ages simplified from the Geologic Map of North America (Reed et al. 2005, Geological Society of America DNAG), and the "ancient core" idea from Hoffman (1988), Annual Review of Earth and Planetary Sciences 16:543-603. Distances are approximate.

UNIT 3.2 · LESSON 4 · EARTH TIMELINE

45 cm Earth timeline: event table and three strips

Use: Glue under "45 cm Earth timeline" (D3-B); cut and tape the strips end to end · Copies: One per student

Lab B. Left end: Earth forms. Right end: today. Scale: 1 cm = 100 million years. Position = (4,500 minus age in millions of years) ÷ 100. Cut the three 15-cm strips along the dashed lines and tape them end to end. Mark and label each event.

Event	Age (million years ago)	Position (cm)
Earth forms	4,500	0
Oldest mineral (zircon)	4,400	1
Oldest known rock	4,000	
Oldest fossils	3,500	
Oldest ocean floor	200	
Today	0	45

Strip 1: 0 to 15 cm (1 cm per box)

Strip 2: 15 to 30 cm

Strip 3: 30 to 45 cm

Questions: How much of the 45 cm is covered by ocean floor that still exists? _____ Why is the oldest Earth rock younger than Earth? _____ Why are there no fossils that old in the ocean floor? _____

Sources: ocean-floor ages, Müller et al. (2008), Science 319:1357-1362, "Age, spreading rates, and spreading asymmetry of the world's ocean crust" (oldest ocean floor about 180 million years; rounded); oldest mineral (zircon), Wilde et al. (2001), Nature 409:175-178 (about 4.4 billion years, Jack Hills, Western Australia); oldest known rock, Bowring and Williams (1999), Contributions to Mineralogy and Petrology 134:3-16 (Acasta Gneiss, Canada, about 4.0 billion years); oldest fossils, Allwood et al. (2006), Nature 441:714-718 (about 3.5 billion years, Western Australia).

UNIT 3.2 · LESSON 4 · EVIDENCE FOR EARLY EARTH

Early Earth evidence log

Use: Glue under "Evidence list" in the notebook · Copies: One per student

List the evidence scientists use to learn about early Earth. Use today's timeline and your Lesson 4 reading (Lesson 8.2 Life's Early History, pp. 219-225).

Evidence	Where it is found	Age	What it tells us

Why can't we use only Earth rocks to learn about the very first few hundred million years? _____

_____

_____

Crater counts: record sheet

Use: Glue under "Crater counts" in the notebook (D3-A) · Copies: One per student

Simulation A. Watch four time steps on the slides: 1, 2, 3, and 4 billion years. At each step count the craters on the Moon and on Earth. Record both numbers.

Time step	Moon craters	Earth craters
1 billion years		
2 billion years		
3 billion years		
4 billion years		

1. Which surface looks older? How can you tell? _____

_____

_____

2. Earth was hit as often as the Moon. Why does Earth show so few craters? _____

_____

_____

3. A surface has a lot of craters. Is it old or young? Why? _____

_____

_____

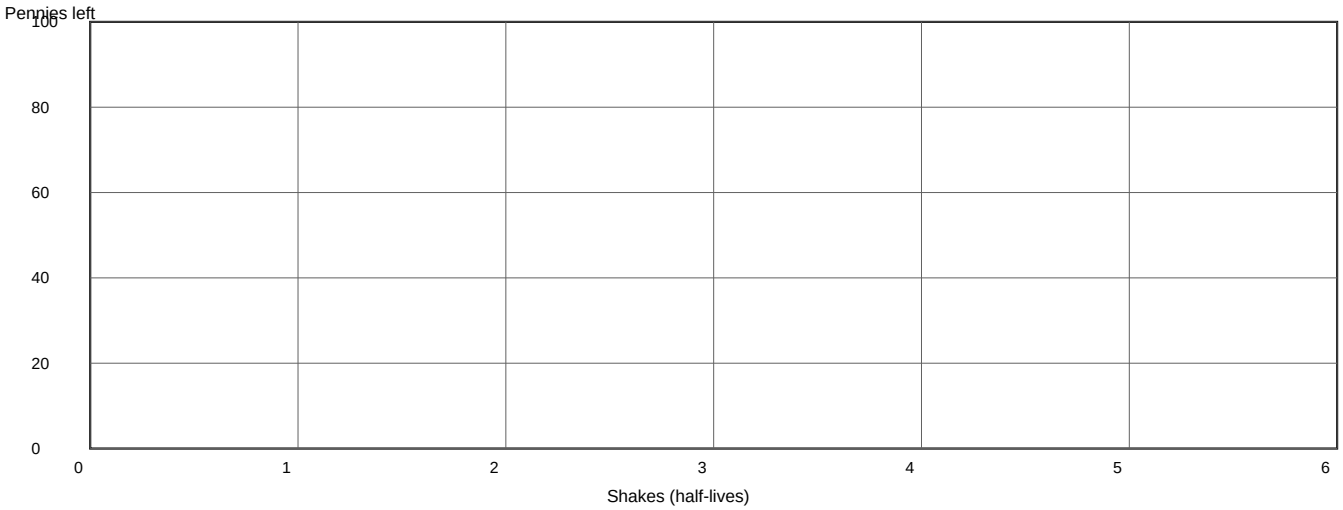
Illustrative data: simplified numbers for class, not from a single study. The Moon has no plate tectonics and almost no weathering, so craters stay; Earth's craters are erased by erosion, lava and plate motion.

Half-life model: decay record

Use: Quick lab (textbook p. 215); glue under "Half-life" in the notebook · Copies: One per pair

Model decay with 100 pennies in a shoebox. Shake the box and pour the pennies out. Remove the pennies that landed heads (they have "decayed"). Count and record the pennies left. Repeat six times. Each shake is one half-life.

Shake (half-lives)	Pennies left (atoms not decayed)	Predicted (halve each time)
0	100	100
1		50
2		25
3		
4		
5		
6		



If a half-life is 700 million years, how old is a rock that has 25% of its original parent atoms left? _____

Why do scientists date meteorites and moon rocks, not Earth's earliest rocks, to find the age of the solar system?

_____

A coin-flip model of radioactive decay (a standard classroom model). Uranium-lead dating of meteorites gives about 4.57 billion years: Connelly et al. (2012), Science 338:651-655. Moon highland (ferroan anorthosite) rock about 4.36 billion years: Borg et al. (2011), Nature 477:70-72.

UNIT 3.2 · LESSON 6 · GUIDED NOTES

Guided notes: the life of ocean crust and how continents grow

Use: Glue under "Guided notes" in the notebook (D4-A) · Copies: One per student

The life of ocean crust

Ocean crust is made at a _____ and moves away about 25 km per million years.

Travel 3,000 km: 3,000 ÷ 25 = _____ million years.

In 200 million years it travels 200 × 25 = _____ km. Then it reaches a trench and sinks.

No ocean floor is older than about _____ million years.

Density and continents

Ocean crust: about 3.0 g/cm³. Continental crust: about 2.7 g/cm³. The _____ plate sinks (subduction).

Each continent has an ancient core. Volcanic arcs and pieces of crust are added to its edge, like tree rings. The edge is _____ than the core.

Densities are typical values (oceanic crust 2.9 to 3.0 g/cm³; continental crust about 2.7 g/cm³). The cooler, older ocean plate is denser than the mantle below it, which is why it can sink.

Density lab table

Use: Glue under "Density lab table" in the notebook (D4-B) · Copies: One per student

Lab. Predict: which block sinks, and which floats? Measure the mass of each block and l × w × h for the volume. Density = mass ÷ volume. Water is 1.0 g/cm³.

Block	Mass (g)	Volume (cm³)	Density (g/cm³)	Sinks or floats?
Wood (continent)				
Clay (old, cold ocean plate)				

Which block is like the ocean plate? _____ What does the denser block do when they meet? _____ What stands in for the mantle? _____

Draw what you saw

Push the blocks toward each other on water. Sketch the side view.

UNIT 3.2 · LESSON 6 · SKETCHNOTES

SketchNotes checklist

Use: Glue above your half page of notes (D4-C) · Copies: One per student

Half page of notes: the life of ocean crust and how continents grow. One page of sensemaking: a cross-section. Check each part you drew.

Part of the cross-section	Done?
Ridge (new crust)	
Plate moving away from the ridge	
Trench	
Volcanic arc	
Continent with an ancient core and a younger edge	
Density labels (3.0 and 2.7 g/cm³)	
Arrow for plankton-shell carbon riding the plate down	

Turn in a photo on Canvas: Lesson 6: SketchNotes.

Magnetic stripes: band table and paper strips

Use: Day 1. Glue under "Magnetic stripes" (D5-A); cut and fold the strips · Copies: One per pair

Lab A (2 cm = 1 million years). Color the two strips with the table. Start at the end you push first (the oldest crust). Cut a slit in a folded sheet of paper (the ridge). Put the strips back to back, push them up through the slit and pull them apart. Tape the result in your notebook. Label the youngest and oldest crust.

Band	Color	Length	Age it represents
1	Reversed (blue)	3.4 cm	about 3.6 to 5.3 million years
2	Normal (red)	2.0 cm	about 2.6 to 3.6 million years
3	Reversed (blue)	3.6 cm	about 0.8 to 2.6 million years
4	Normal (red)	1.6 cm	0 to 0.8 million years (youngest)

Strip A (cut out; color bands 1 to 4, 10.6 cm total)

Strip B (cut out; color the same bands)

What does a pair of mirror-image stripe patterns tells you about how the crust formed? _____

Polarity timescale (Gilbert reversed, Gauss normal, Matuyama reversed, Brunhes normal): Gradstein et al. (2012), The Geologic Time Scale 2012; first recognized in ocean-floor data by Vine and Matthews (1963), Nature 199:947-949. Band lengths use 2 cm = 1 million years and are rounded.

Hawaii and GPS: measured motion

Use: Day 1. Glue under "Hawaii and GPS" (D5-B) · Copies: One per student

Lab B. Rate = distance ÷ time. 1 km per million years = 0.1 cm per year. Find the rate for Kauai and for Midway. GPS stations show about 7 to 8 cm per year. What do you notice?

Island	Distance from Kilauea (km)	Age (million years)	Rate (cm per year)
Kilauea	0	0	(starting point)
Kaua'i	about 500	about 5	
Midway Atoll	about 2,500	about 28	

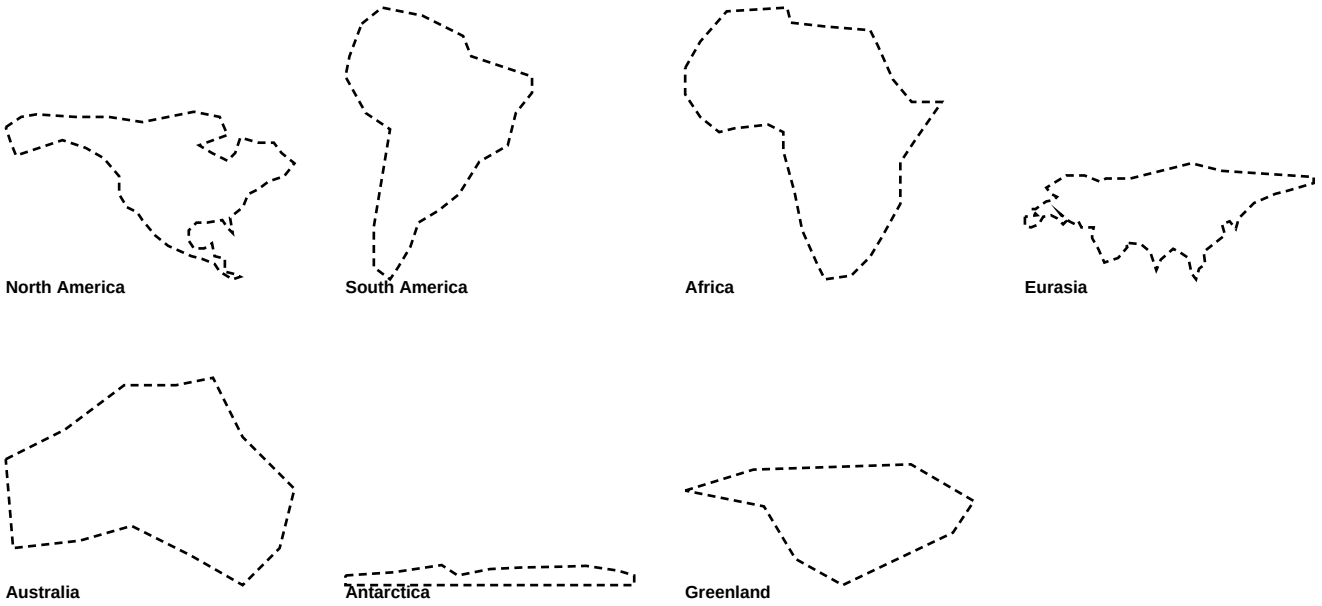
What do you notice about the rates and about GPS? _____

Island ages and distances rounded from the U.S. Geological Survey, "The Hawaiian-Emperor chain" and Clague and Dalrymple (1987), USGS Professional Paper 1350. GPS motion of the Pacific plate about 7 to 8 cm per year (NASA/JPL and UNAVCO GPS velocity solutions; rounded).

Pangaea puzzle: continent cutouts and fossil table

Use: Day 2. Cut out the continents; glue the fossil table under "Pangaea puzzle" (D5-C) · Copies: One per pair

Lab C. Cut out the dashed continents. Fit them together like a puzzle and tape the best fit in your notebook. Mark where each fossil lived.



Fossil	What it was	Found in
Mesosaurus	freshwater reptile	South America and southern Africa
Glossopteris	seed fern (heavy seeds)	South America, Africa, India, Australia, Antarctica
Lystrosaurus	land animal	Africa, India, Antarctica

Mesosaurus lived in fresh water and Glossopteris had heavy seeds. Why does that matter? _____ What can fossils NOT tell you that GPS can? _____

Continent outlines are simplified schematic shapes and are not to scale. Fossil distributions: classic evidence summarized in Wegener (1915/1929) and in standard textbooks.

Evaluate the evidence table

Use: Day 2. Glue under "Evaluate the evidence" (D5-D) · Copies: One per student

Rate each line of evidence for how strongly it supports moving crust (1 = weak, 4 = very strong).

Line of evidence	Measured?	Past or present?	Independent?	Strength (1 to 4)
Rock ages across a ridge				
Magnetic stripes				
GPS and Hawaii				
Fossils and rocks				

Which is strongest for current motion? _____ For past motion? _____ Why does agreement of two independent methods help? _____

_____

_____

Three buckets: sort your evidence

Use: Day 1. Glue under "Sort your evidence" (D6-E) · Copies: One per student

Put each row of your Unit CER log in a bucket. Then look back at your Lesson 1 claim: what would you change now?

The seafloor is young because...	The continents are old because...	The early record is missing because...

What I would change in my Lesson 1 claim: _____

UNIT 3.2 · LESSON 8 · PART A: ARGUMENT

Part A sentence frames and data bank

Use: Day 1. Glue under "Part A: argument" (D6-A) · Copies: One per student

Claim: Plate tectonics explains the ages (very well / partly / poorly) because _____.

Ocean evidence: The seafloor is youngest at the _____ and gets older with _____ ; the oldest ocean floor is about _____ million years old.

Continent evidence: Continental crust as old as _____ exists. In North America the crust gets _____ toward the edges.

Reasoning: Because ocean plates are _____ than continental crust, they _____ at subduction zones, while continents _____ .

Limit: One thing this evidence cannot show is _____ . OR: Someone might say _____ . That does not fit the evidence because _____ .

Data you may use

Crust age: 0 to 120 million years across 3,000 km (2.5 cm per year). Oldest ocean floor: about 200 million years. Oldest continental mineral: about 4.4 billion years. North America: 2.7 billion years at the core, 400 million years in the Appalachians. Ocean crust 3.0 g/cm³. Continental crust 2.7 g/cm³. GPS 2 cm per year (spreading rate) and rock ages 2.5 cm per year. Oldest fossils 3.5 billion years, in continental rock.

Values are the unit data from Lessons 1 to 7 (see sources on those sheets).

Star and step peer feedback

Use: Day 1. Glue under "Peer feedback" (D6-D) · Copies: One per student

Trade papers. Check the four questions, then write one star and one step.

Check	Yes / not yet
Does the claim say how well and give a cause?	
Is the evidence specific, from ocean and continents?	
Does the reasoning say why the evidence supports the claim?	
Is a limit or counterclaim addressed?	

Star (something strong): _____

_____

_____

Step (something to add or fix): _____

_____

_____

UNIT 3.2 · LESSON 8 · PART B: EARTH'S STORY

Earth's story: evidence cards and frames

Use: Day 2. Cut out the cards; glue under "Part B" (D6-B, D6-C) · Copies: One per student

Cut out the six evidence cards. Put them in order from oldest to newest on a paper strip and glue them down. Then write the story with the frames.

Meteorites: 4.56 billion years · The solar system formed.	**Moon highlands: up to 4.4 billion years · Solid crust existed early.**
Earth's oldest zircons: 4.4 billion years · Solid crust on Earth by then.	**Oldest Earth rock: 4.0 billion years · Rock survives in the continents.**
Oldest fossils: 3.5 billion years · Life existed by then.	**Oldest ocean floor: 200 million years · Ocean crust is recycled.**

Story frames: About ___ billion years ago, ___ happened. The evidence is ___ .

_____

_____

_____

_____

Why is Earth's earliest record missing? _____

_____

_____

Sources: Connelly et al. (2012), Science 338:651-655 (meteorites, 4.567 billion years); Borg et al. (2011), Nature 477:70-72 (moon highlands); Wilde et al. (2001), Nature 409:175-178 (zircons); Bowring and Williams (1999) (oldest rock); Allwood et al. (2006), Nature 441:714-718 (fossils); Müller et al. (2008), Science 319:1357-1362 (ocean floor). Ages rounded.

UNIT 3.2 · LESSON 9 · REVIEW DAY

Self-rating chart and work plan

Use: Glue under "Self-rating chart" in the notebook (D7-A) · Copies: One per student

Rate yourself 1 to 4 (1 = I need help, 4 = I could teach it). Then list what you will do today: unfinished CFA, review, or enrichment.

Target	Rating (1 to 4)	CFA score
ESS1-5.1 Plates, boundaries, and crust		
ESS1-5.2 Reading rock ages		
ESS1-5.3 Why ocean crust is young and continents are old		
ESS1-5.4 Evidence for moving crust		
ESS1-5.5 How well plate tectonics explains rock ages		
ESS1-6.1 Evidence for early Earth		
ESS1-6.2 Why meteorites and moon rocks give the age		
ESS1-6.3 Telling the story of Earth's formation		

My work plan for today

UNIT 3.2 · LESSON 9 · REVIEW STATIONS

Review stations: five station cards

Use: Cut apart; early finishers rotate through the stations · Copies: One set per station table

Station tasks, one for each priority target. Do the task in your notebook, then check with the answer card at the station.

Station 1 · ESS1-5.1: Sketch a ridge, a trench and a transform fault. Label where ocean crust is made and where it sinks.

Station 2 · ESS1-5.2: Crust at 1,500 km from a ridge is 60 million years old; at 2,500 km it is 100 million. Describe the pattern and the rate in cm per year.

Station 3 · ESS1-5.3: Explain in two sentences why ocean floor is under 200 million years old but continents hold 4 billion-year-old rock.

Station 4 · ESS1-5.4: Rank magnetic stripes, GPS, and fossils for how strongly they show crust moves. Give one reason for the top rank.

Station 5 · ESS1-5.5: Write a claim that plate tectonics explains rock ages, with two numbers as evidence and one limit.

Data are from the unit; no new sources.